

CINCO DE CHILDREN

Abi Mangual

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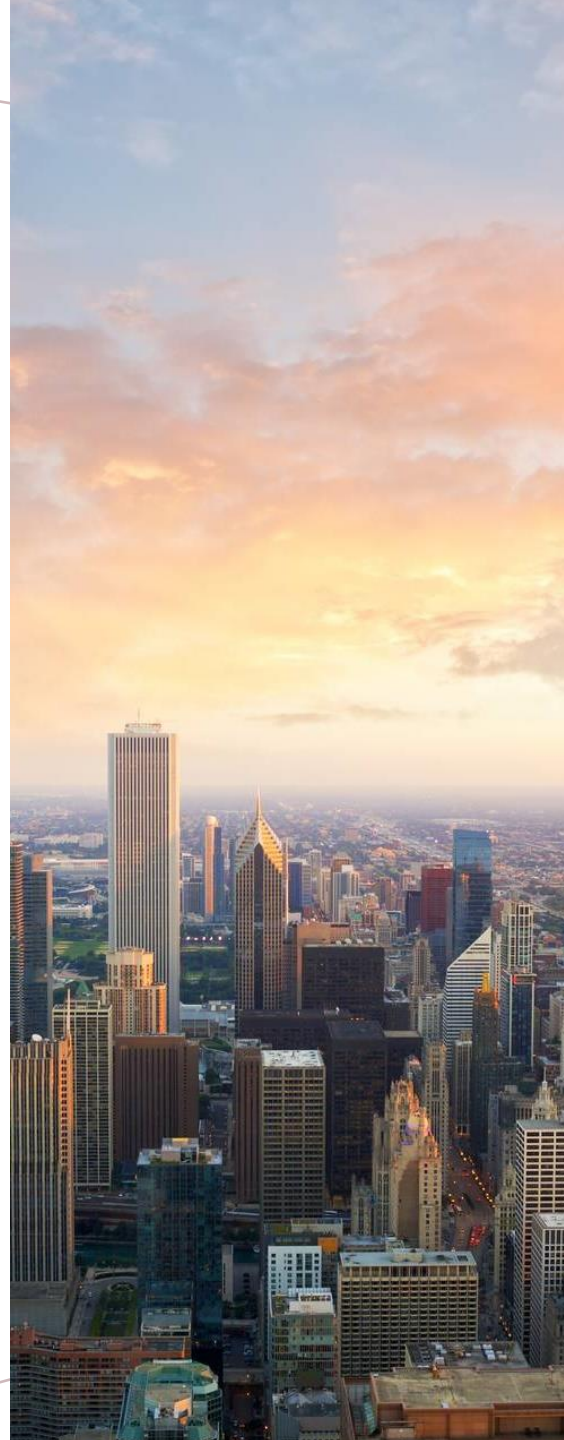
Michael Carter

Introduction & Background

- The education inequality and education gaps that exist within impoverished groups are considered one of the most important determinants of social mobility in the United States.

- One of the most prominent causes for education inequality is poverty or more specifically child poverty among ages 5 through 17.

- The age range of 5 through 17 are widely considered to be one of if not the most important phases of human development, yet students raised under economic scarcity are consistently shown to have worse learning outcomes due to factors outside of their control.

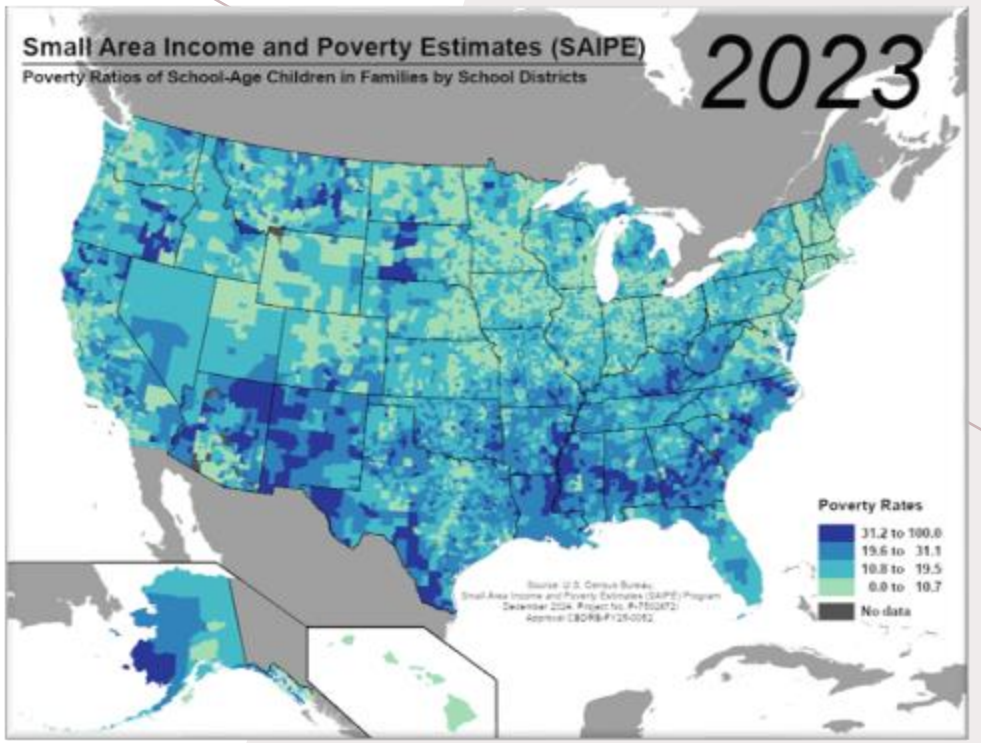


RESEARCH QUESTION

WHAT IS THE RELATIONSHIP BETWEEN POVERTY AND EDUCATION?

DATASETS

- PRIMARY DATASET WAS THE SEDA TEST SCORES DATASET ACQUIRED FROM THE STANFORD EDUCATION DATA ARCHIVE. USES TEST SCORE DATA FROM EACH DISTRICT
- MERGED WITH COVARIATE DATA WHICH INCLUDES IMPORTANT DEMOGRAPHICAL VARIABLES
- CROSSWALK DATASET THAT INCLUDES LEAID WHICH IS THE UNIQUE IDENTIFIER USED FOR MERGING
- AND FINALLY THE SMALL AREA INCOME AND POVERTY ESTIMATES. POVERTY RATES FOR STUDENTS WITHIN SCHOOL DISTRICTS.





METHOD

Dependent Variables:

- Test Scores (cs_mn_all)

Independent Variables:

- Poverty Rate (SAIPE)
- SESAVG (Socioeconomic Status average 2005-09, 2010-14, & 2015),
- SESAVGALL (to account for the lack of inputs in the year 2024)

Control Variables:

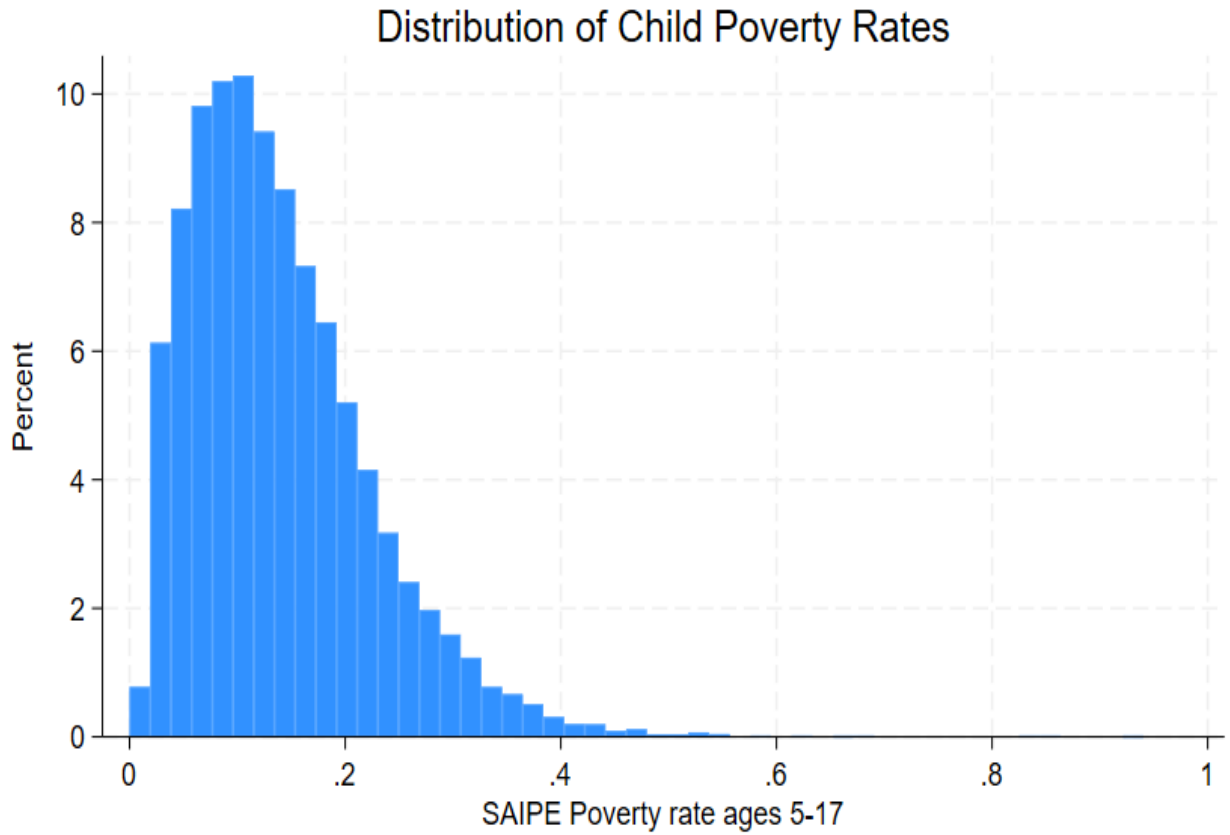
- Locale (suburban, town, & rural)
- State Fixed Effects (absorb(fips) vce(cluster fips)
- Enrollment (totenrl)

STRATEGY:

1. Merge SEDA With Covariate, Crosswalk, and SAIPE.
2. Sample Size: 97,000 Observations
3. Square poverty to capture diminishing marginal harm
4. Poverty captures current stress while SESAVGALL captures long-term structural support.
5. Use state-fixed effects to isolate state variation and varying educational standards.

SAIPE POVERTY RATE:

- Majority of districts sit in lower end of poverty
- Right-skewed distribution.
- Significant long tail of districts facing economic hardship.
- Long tail districts acted as a weight, pulling regression results out of alignment.
- Used a quadratic model to adjust for this.



REGRESSION MODEL:

$$Y_{ids} = \beta_0 + \beta_1 \text{Poverty}_{ds} + \beta_2 \text{Poverty}_{ds}^2 + \beta_3 \text{SES}_{ds} + \beta_4 \text{Enrl}_{ds} + \mathbf{L}_{ds}\Gamma + \alpha_s + \epsilon_{ids}$$

Stata Code

```
"areg cs_mn_all est_population_5_17_poverty_pct poverty_sq sesavgall totenrl suburb town rural,
 absorb(fips) vce(cluster fips)"
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- Beta 1 and 2 adjust for the nonlinearity of poverty rate. Beta 2 represents the curvature and rate of change of the slope.
- Beta 3 showcases the effect of long-term structural environments on test scores.
- Beta 4 controls for district size and resources
- L stands for locale and T stands for what area (Rural, Town, Suburb), using Urban as base.
- Alpha represents the state fixed effects and controls for different state tests or funding policies.

REGRESSION RESULTS:

Linear regression, absorbing indicators

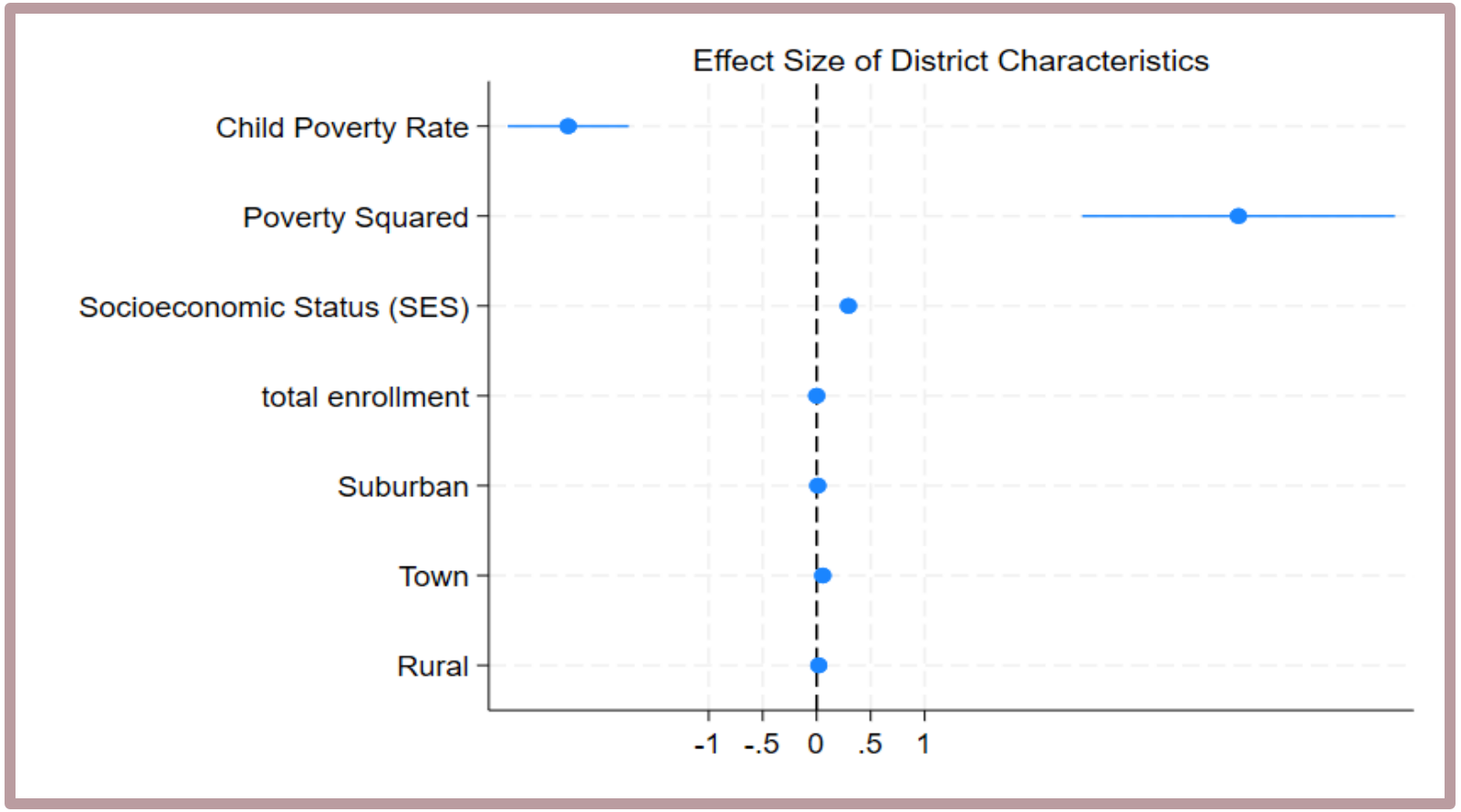
Absorbed variable: **fips**

Number of obs	= 97,977
No. of categories	= 43
F(7, 42)	= 162.02
Prob > F	= 0.0000
R-squared	= 0.4866
Adj R-squared	= 0.4863
Root MSE	= 0.3381

(Std. err. adjusted for 43 clusters in **fips**)

cs_mn_all	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
est_population_5_17_poverty_pct	-2.301907	.2779158	-8.28	0.000	-2.862764	-1.741051
poverty_sq	3.907525	.7177951	5.44	0.000	2.458956	5.356094
sesavgall	.2943507	.0207863	14.16	0.000	.2524022	.3362991
totenrl	-8.51e-07	3.55e-07	-2.40	0.021	-1.57e-06	-1.35e-07
suburb	.0110844	.0137059	0.81	0.423	-.0165753	.0387441
town	.0571941	.0257362	2.22	0.032	.0052562	.1091319
rural	.019934	.0203963	0.98	0.334	-.0212273	.0610954
_cons	.0129114	.031142	0.41	0.681	-.0499357	.0757585

MODEL CONTINUED:



Key Points

- Socioeconomic Status is the largest positive coefficient.
- Poverty has a large effect on test scores
- Poverty Squared is positive which showcases how test scores begin to flatten out as poverty rates increase or "go deeper."
- Students tend to outperform their urban baseline counterpart.

KEY TAKEAWAYS

What is the relationship between poverty and education?

- There is a negative and statistically significant relationship between poverty and test scores.
- Higher poverty is associated with lower academic achievement.
- SESAVG showcases long-term affects on grades within these districts.
- Districts entering poverty are dramatically affected by test scores but districts that have long been in poverty begin to recover grade wise.
- Conclusion: poverty significantly ***reduces*** academic achievement

CONCLUSION

- There is strong and consistent evidence that higher poverty levels are associated with lower student achievement across districts.
- An increase in poverty from 10% to 20% is linked to about a 0.24-point decline in average test scores when no controls are included.
- This negative relationship remains statistically significant even after controlling for factors like socioeconomic status (SES) and total enrollment, though the size of the effect becomes smaller.
- Overall findings suggest that poverty reduces access to educational resources and opportunities, leading to poorer academic outcomes.
- Even if a family is within poverty, SES still gives those students a chance to raise test scores.
- Policy funding and programs can help mitigate the effects from poverty.

FINAL EMPIRICAL ANALYSIS RESULTS

- Test scores decline sharply as districts first enter poverty
- The largest negative impact occurs at lower to moderate poverty levels
- As poverty increases further, the decline in scores begins to level off or slightly improve
- This pattern suggests a diminishing marginal effect of poverty on academic achievement



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